

Practical 2: Setting up Version Control System

Version control, also known as source control, is the practice of tracking and managing changes to software code. Version control systems are software tools that help software teams manage changes to source code over time.

Android Studio supports a variety of version control systems (VCSs), including Git, GitHub, CVS, Mercurial, Subversion, and Google Cloud Source Repositories.

1. Download and install Git. (<https://git-scm.com/downloads>)
2. Open the **Git Command** or **Windows Command-prompt** or **Android Studio command prompt**, enter the following commands:

```
git config --global user.name "Your GitHub ID"
git config --global user.email "Your Email Address"
```

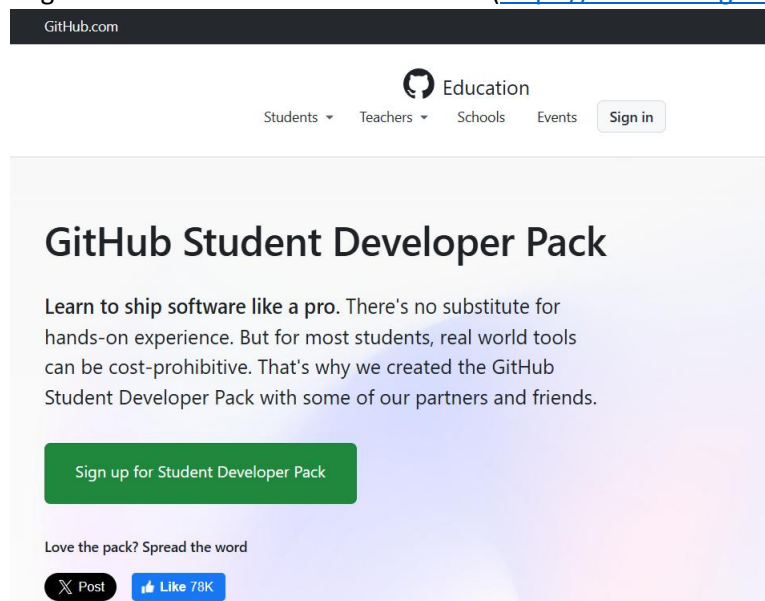
Confirm that you have set the Git username correctly, enter the command:

```
git config --global user.name
git config --global user.email
```

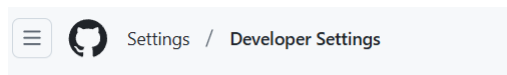
And you shall see your name:

```
> Your name here
> Your email here
```

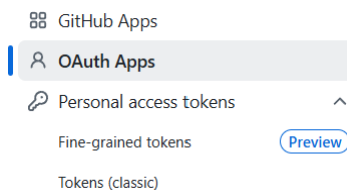
3. Register a new GitHub student account. (<https://education.github.com/pack>)



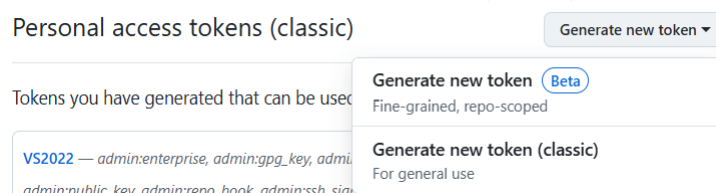
4. Login to your GitHub account. Click on your **profile picture** on the top-right corner, and click **Settings** (⚙️ Settings).
5. On the left panel, click the last item named **Developer settings** (<> Developer settings). You shall see the Developer Settings list as below:



6. Click on the **Personal access tokens**, and select **Tokens (classic)**. Personal access tokens are an alternative to using passwords for authentication to GitHub when using the GitHub API, the command line, or an IDE.



7. On the right of the **Personal access tokens (classic)** panel, click the **Generate new token** button, and select **Generate new token (classic)**.



8. Assign a **Note** (e.g. Android Studio) to the token and set the **Expiration** to **No expiration**.

New personal access token (classic)

Personal access tokens (classic) function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to [authenticate to the API over Basic Authentication](#).

Note

Android Studio

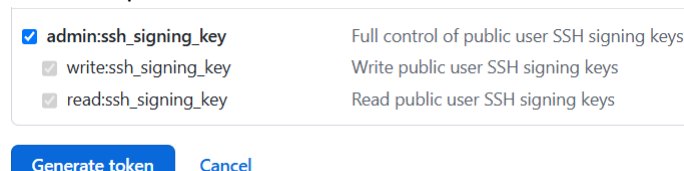
What's this token for?

Expiration *

No expiration

The token will never expire!

9. Select scopes associated with the token and click the **Generate token** button.

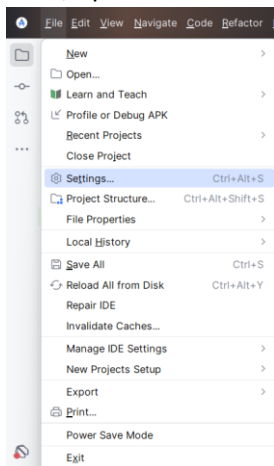


10. Click the **Copy** (📄) button once the token has been generated.

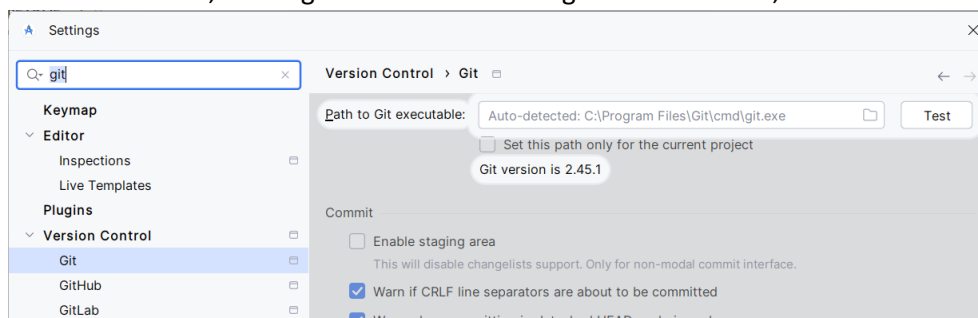
Personal access tokens (classic)

[Generate new token](#)Tokens you have generated that can be used to access the [GitHub API](#). Make sure to copy your personal access token now. You won't be able to see it again!✓ ghp_qIpzfHi9dw4xa4kv6Rw2H4xfHqRfP0KVsxA [Delete](#)

11. Next, open a new Flutter project file. Click on the **File** menu and **Settings**.

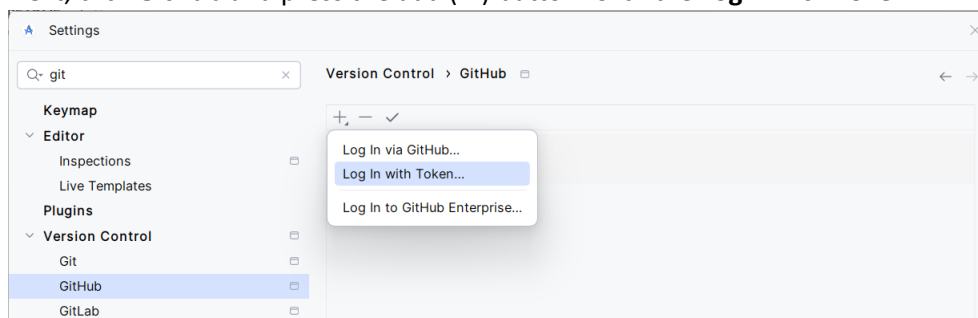


12. In the search bar, enter 'git'. Under the heading **Version Control**, click **Git**.

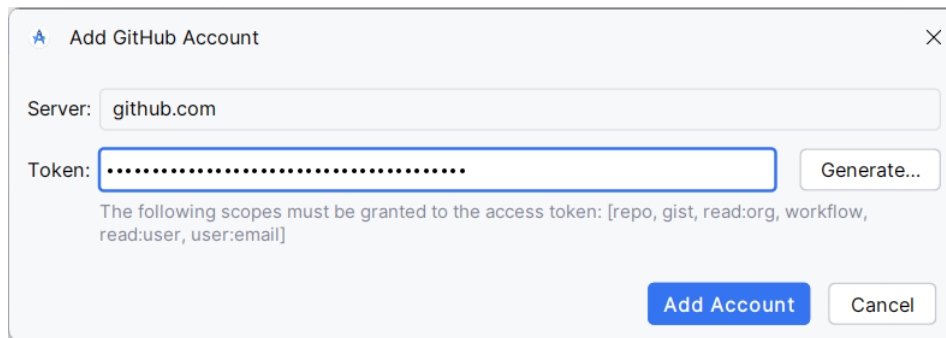


In the **Path to Git executable**, assign the git.exe path and press the **Test** button. You shall see the version code appear below the path.

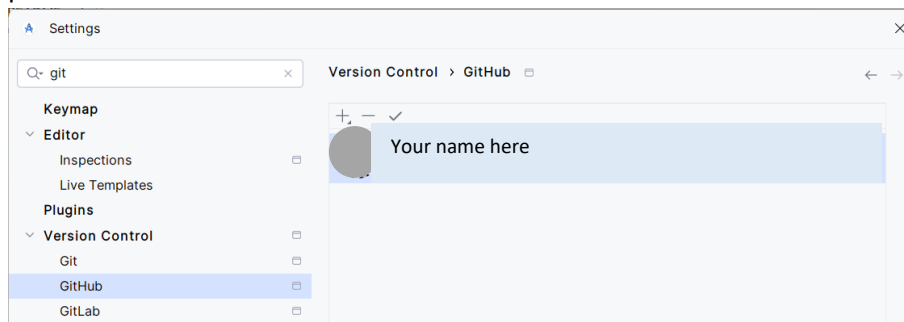
13. Next, click **GitHub** and press the add (+) button. Click the **Login with Token...**



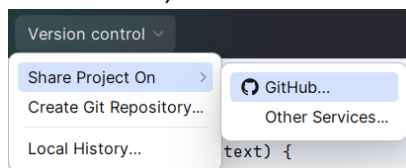
14. Paste the newly token to the Add GitHub Account screen and click the **Add Account** button.



15. Once completed, you should be able to see your profile appear in the list. Close the Settings panel.



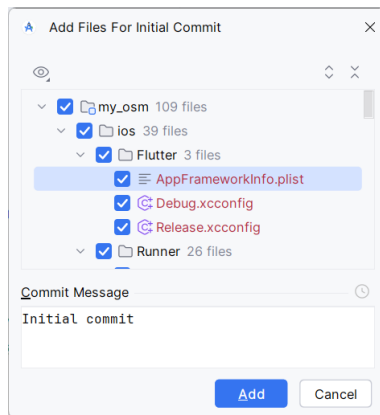
16. On the menu, click **Version Control** and select **Share Project On -> GitHub...**



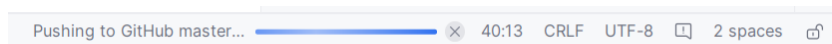
17. When prompted, enter a repository name, set the visibility (Private or Public), and also enter a short description about the project. Click the **Share** button to continue.



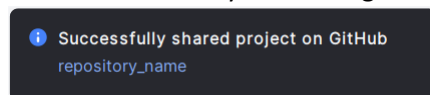
18. You will be prompted to add files for the initial commit. Review files to be added to the repository and press the **Add** button to continue.



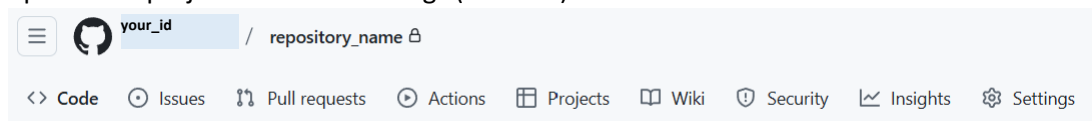
19. At the bottom right of the screen, you can observe the progress of pushing the files to the GitHub cloud.



20. If everything goes well, you will see a message indicating the project has been shared on GitHub successfully. Click the given link to view the project files.

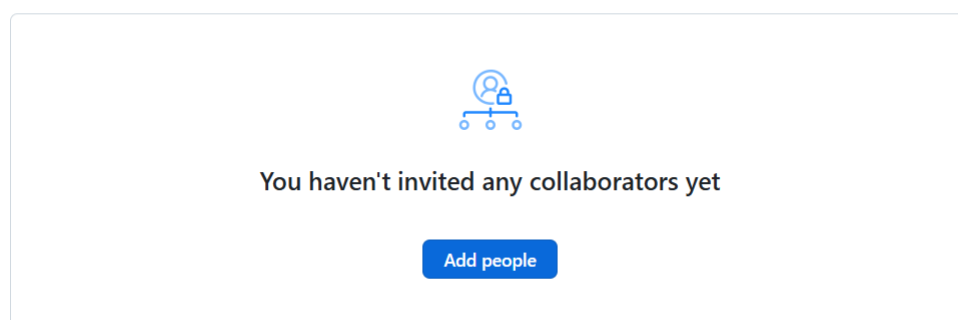


21. Your project will be shown in a browser. You may invite your team members to view, edit and update the project. Click the Settings (⚙️ Settings).

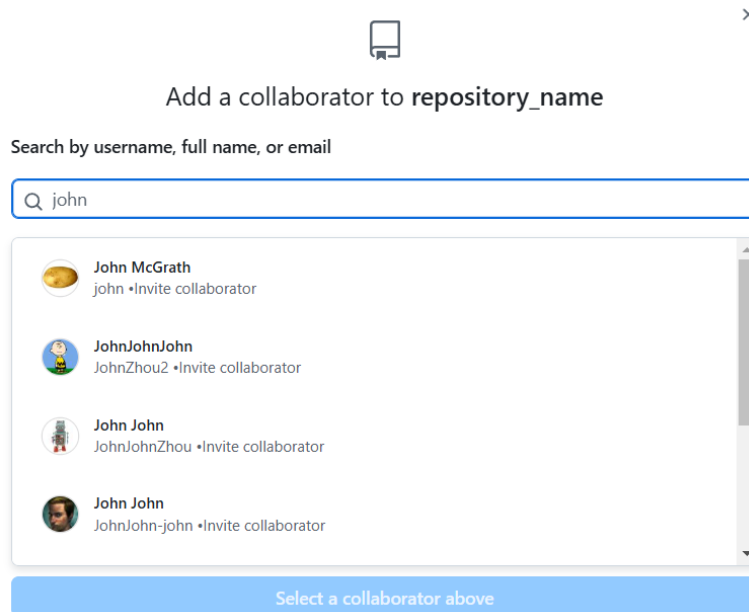


22. Select Collaborators (👤 Collaborators). Under the heading **Manage access**, click the **Add people** (Add people) button.

Manage access

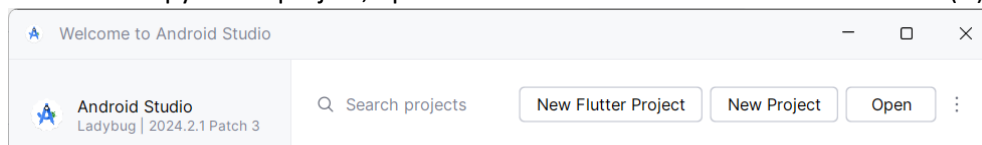


Enter IDs of your team member in the space provided:

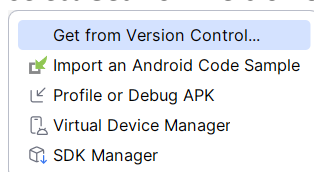


Click Select a collaborator from the list and click **Add [Team Member Name] to this repository**.

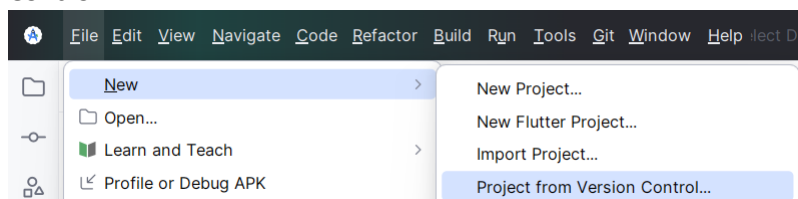
23. You may also share your repository to a member who would like to view your repository, however, not able to edit or update the repository. Click the **<> Code** tab and then the **Code** (**<> Code**) button to show the link (with extension .git) for the project. Click the **Copy** (📄) button next to the clone link. You may share the git link to a member who would like to clone a copy of your project.
24. To clone a copy of the project, open Android Studio. Click on the More Actions (⋮) button.



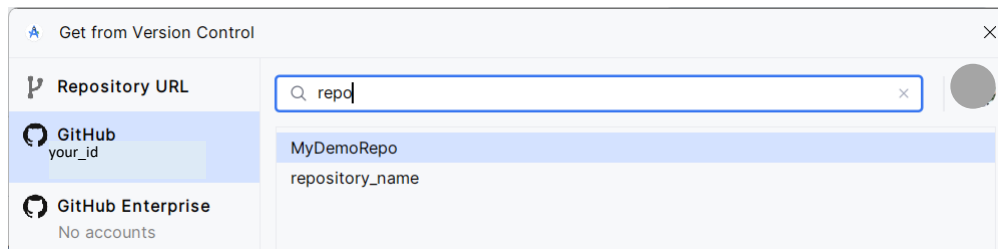
Select **Get from Version Control...**



Or, if you have opened an existing project, click **File -> New -> Project From Version Control...**

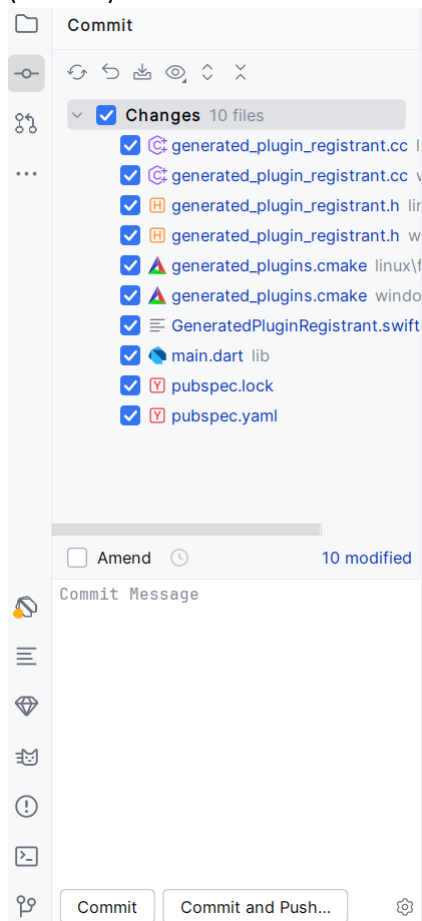


25. On the left panel, select your GitHub account and enter a search key (E.g. repo). Your project will appear in the repository list. Press the Clone (**Clone**) button to proceed.



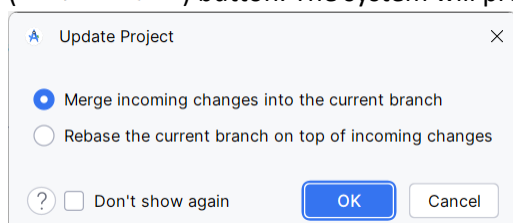
That's all you need to create and clone a repository.

26. After cloning a project, you may make some changes to the project files/codes/etc. Once you have completed all the necessary changes, you must **commit** the project. Click the VCS (main) button and then Commit (Commit...) button. Check all the files in the Commit panel.



Enter a commit message and click the **Commit and Push** button.

To view changes made by other team members, on the VCS menu, click the Update Project (Update Project...) button. The system will prompt you to Update the project:





TODO: Visit <https://docs.github.com/en/get-started/start-your-journey/hello-world> and learn GitHub essentials!